WORKED SOLUTION

STEP III 2006, Question 2

Pure • Integration

STEP III 2006, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q2

The key idea

The interval $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$ is symmetric, so replacing θ by $-\theta$ gives a second expression for I; adding the two clears the awkward $\sin\theta$ from the denominator. Once everything is written in terms of $\tan\theta$, the integrals I and J have the same denominator, and a well-chosen combination of them has integrand exactly 1. Throughout, keep an eye on the sign of $\cos 2\alpha$: it is what changes in the final part.

Solution

Write $s = \sin 2\alpha$ and $c = \cos 2\alpha$, so $s^2 + c^2 = 1$. In both ranges of α considered ($\alpha \neq \frac{1}{4}\pi$) we have $0 < s < 1$, so $1 \mp s\sin\theta \geq 1 - s > 0$: the integrand of I is continuous on the whole interval.

Part (i)

Substitute $\theta = -\phi$, so $\mathrm{d}\theta = -\mathrm{d}\phi$ and the limits swap. Since $\cos^2(-\phi) = \cos^2\phi$ and $\sin(-\phi) = -\sin\phi$,

$$I = \int_{\frac{1}{2}\pi}^{-\frac{1}{2}\pi} \frac{\cos^2\phi}{1 + \sin\phi\, s}\,(-\mathrm{d}\phi) = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\cos^2\phi}{1 + s\sin\phi}\,\mathrm{d}\phi,$$

which is the first result (renaming ϕ as θ).

Adding the two expressions for I,

$$2I = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \cos^2\theta \left(\frac{1}{1 - s\sin\theta} + \frac{1}{1 + s\sin\theta}\right)\mathrm{d}\theta = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{2\cos^2\theta}{1 - s^2\sin^2\theta}\,\mathrm{d}\theta.$$

Now divide top and bottom by $\cos^2\theta$ (non-zero inside the interval):

$$\frac{2\cos^2\theta}{1 - s^2\sin^2\theta} = \frac{2}{\sec^2\theta - s^2\tan^2\theta} = \frac{2}{1 + \tan^2\theta\,(1 - s^2)} = \frac{2}{1 + c^2\tan^2\theta},$$

using $\sec^2\theta = 1 + \tan^2\theta$ and $1 - \sin^2 2\alpha = \cos^2 2\alpha$. Hence

$$2I = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{2}{1 + \tan^2\theta \cos^2 2\alpha}\,\mathrm{d}\theta,$$

as required. (At $\theta = \pm\frac{1}{2}\pi$ this integrand tends to 0, so nothing goes wrong at the ends.)

Part (ii)

The $\sec^2\theta$ in the numerator invites $t = \tan\theta$, $\mathrm{d}t = \sec^2\theta\,\mathrm{d}\theta$. As θ runs from $-\frac{1}{2}\pi$ to $\frac{1}{2}\pi$, t runs over the whole real line:

$$J = \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{1 + c^2t^2}.$$

Since $0 < \alpha < \frac{1}{4}\pi$, we have $0 < 2\alpha < \frac{1}{2}\pi$ and so $c > 0$. Putting $u = ct$,

$$J = \frac{1}{c}\int_{-\infty}^{\infty} \frac{\mathrm{d}u}{1 + u^2} = \frac{1}{c}\Big[\arctan u\Big]_{-\infty}^{\infty} = \frac{\pi}{c} = \pi\sec 2\alpha.$$

Part (iii)

By part (i), $I = \displaystyle\int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1 + c^2\tan^2\theta}$, which has the same denominator as J. So

$$s^2I + c^2J = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{s^2 + c^2\sec^2\theta}{1 + c^2\tan^2\theta}\,\mathrm{d}\theta.$$

? **Ask yourself** Why should this particular combination be any easier?

Look at the numerator: $s^2 + c^2\sec^2\theta = s^2 + c^2 + c^2\tan^2\theta = 1 + c^2\tan^2\theta$. It is exactly the denominator, so the integrand is 1 and

$$s^2I + c^2J = \pi.$$

Using $J = \pi/c$ from part (ii),

$$s^2I = \pi - \pi c = \pi(1 - c) \quad\Longrightarrow\quad I = \frac{\pi(1-c)}{1 - c^2} = \frac{\pi}{1 + c},$$

where we used $s^2 = 1 - c^2 = (1 - c)(1 + c)$ and cancelled $1 - c \neq 0$ (as $c < 1$). Finally $1 + \cos 2\alpha = 2\cos^2\alpha$, so

$$I = \frac{\pi}{2\cos^2\alpha} = \tfrac{1}{2}\pi\sec^2\alpha. \qquad \checkmark$$

Part (iv)

Now $\frac{1}{4}\pi < \alpha < \frac{1}{2}\pi$, so $\frac{1}{2}\pi < 2\alpha < \pi$: still $0 < s < 1$, but now $c < 0$. Part (i) used only $s^2 + c^2 = 1$, so it still holds, and so does the identity $s^2I + c^2J = \pi$ from part (iii). The one thing that changes is part (ii).

Ask yourself Where exactly did part (ii) use $\alpha < \frac{1}{4}\pi$?

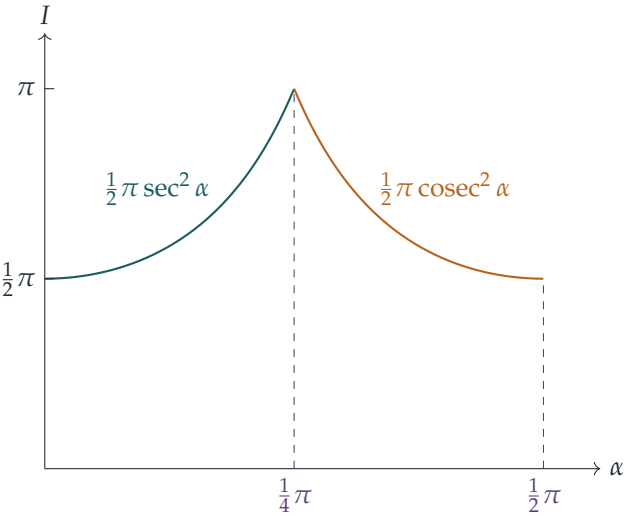
In the substitution $u = ct$: if $c < 0$ the limits of u come out as $+\infty$ to $-\infty$, and swapping them back gives a minus sign. In other words, $J = \pi/|c| = -\pi/c$. (This must be so: J is the integral of a positive function, so it cannot equal the negative number $\pi \sec 2\alpha$.) Then

$$s^2 I = \pi - c^2\Big(-\frac{\pi}{c}\Big) = \pi(1+c) \quad \Longrightarrow \quad I = \frac{\pi(1+c)}{(1-c)(1+c)} = \frac{\pi}{1-c},$$

cancelling $1 + c \neq 0$ (as $c > -1$). Since $1 - \cos 2\alpha = 2\sin^2\alpha$,

$$I = \frac{\pi}{2\sin^2\alpha} = \tfrac{1}{2}\pi \operatorname{cosec}^2\alpha \qquad \big(\tfrac{1}{4}\pi < \alpha < \tfrac{1}{2}\pi\big).$$

There is a quick confirmation: I depends on α only through $\sin 2\alpha$, and $\sin 2(\frac{1}{2}\pi - \alpha) = \sin 2\alpha$. So $I(\alpha) = I(\frac{1}{2}\pi - \alpha) = \frac{1}{2}\pi \sec^2(\frac{1}{2}\pi - \alpha) = \frac{1}{2}\pi \operatorname{cosec}^2\alpha$. The graph of I is symmetric about $\alpha = \frac{1}{4}\pi$, where both formulas give π:



Checks

- At $\alpha = \frac{1}{4}\pi$ ($c = 0$) the 2I-form of part (i) has integrand 2, giving $I = \pi$, matching both formulas.
- As $\alpha \to 0$, $I \to \int \cos^2\theta \, d\theta = \frac{1}{2}\pi$, matching $\frac{1}{2}\pi \sec^2 0$.
- Numerically (Simpson's rule on the original integral), $\alpha = 0.5$ gives $I \approx 2.039595$, equal to $\frac{1}{2}\pi \sec^2 0.5$; and $\alpha = 1.1$ gives $I \approx 1.977709 = \frac{1}{2}\pi \operatorname{cosec}^2 1.1$. In every case tested, $s^2 I + c^2 J$ came out as 3.141593.

Common slips

- Writing $J = \pi \sec 2\alpha$ in part (iv) as well: this is negative there, which is impossible for the integral of a positive function.
- Forgetting that the substitution $t = \tan\theta$ turns the finite interval into the whole real line, so the limits become $\pm\infty$.

Beyond STEP

The trick in part (i), averaging $f(\theta)$ and $f(-\theta)$, is the start of splitting a function into even and odd parts: only the even part survives integration over a symmetric interval. Integrals of rational functions of $\sin\theta$ and $\cos\theta$ over a full period are a classic application of contour integration (the residue theorem) in a first complex-analysis course, where results like this one drop out in a few lines; see Needham, *Visual Complex Analysis*.

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